Individual Printer Performance and Anomaly Report

EPR-007

Perfect Binder · MullerBind Pro · Plant 2 - Hall B

PRINTER INFORMATION

Printer ID	EPR-007
Printer name	Perfect Binder
Model	MullerBind Pro
Machine type	Binding
Location	Plant 2 - Hall B
Installed	2020-07-16
Last recorded status	IDLE
Status mix over period	IDLE 448, PRINTING 399, ERROR 17
Total telemetry readings	864
Monitoring period	2026-10-15 00:00:00 to 2026-10-17 23:55:00
Report generated	2026-10-26 13:43:38.406

CONFIGURED SENSOR THRESHOLD LIMITS

SENSOR	LOWER LIMIT	UPPER LIMIT	SEVERITY
Temperature (°C)	—	60.0	CRITICAL
Humidity (%)	30.0	70.0	WARNING
Ink level (%)	15.0	—	WARNING
Paper stock (%)	10.0	—	INFO
Motor RPM (rpm)	—	3800.0	WARNING
Vibration (mm/s)	—	6.5	CRITICAL
Power (kW)	—	16.2	WARNING
Paper jams (count)	—	2.0	CRITICAL

SENSOR PERFORMANCE SUMMARY

SENSOR (UNIT)	MIN	AVG	MAX	LAST VALUE	BREACH READINGS
Temperature (°C)	24.9	31.73	39.8	26.2	0
Humidity (%)	36.2	48.46	84.7	53.3	18
Ink level (%)	51.5	66.11	84.0	51.5	0
Paper stock (%)	53.4	71.07	93.1	53.4	0
Motor RPM (rpm)	0.0	1445.14	3119.0	49.0	0
Vibration (mm/s)	0.1	1.47	11.4	0.2	6
Power (kW)	1.0	6.40	12.9	1.4	0
Throughput (ppm)	0.0	27.38	63.0	0.0	0
Paper jams (count)	0.0	0.07	5.0	0.0	9

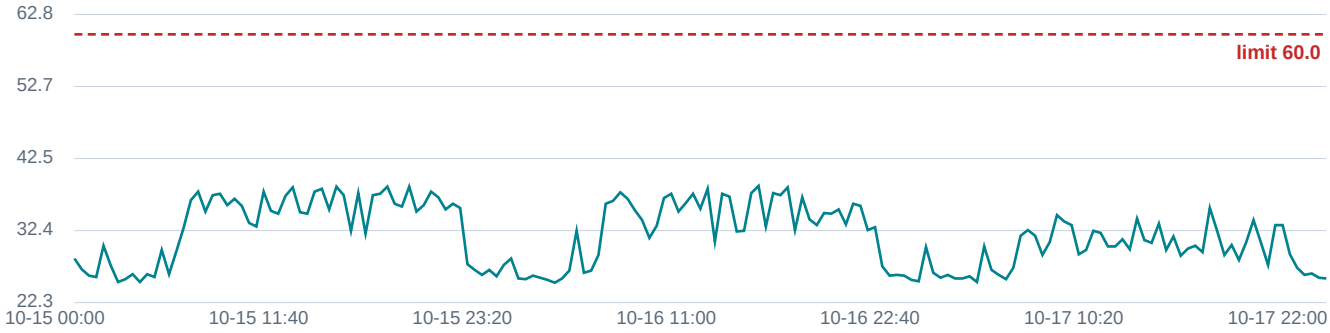
ANOMALY DETECTION SUMMARY

Labelled anomalous readings	35 of 864 readings (4.1%)
Raw threshold-breach readings	27 readings with at least one sensor out of bounds
Anomaly types	PAPER_JAM_BURST: 17 · VIBRATION_SPIKE: 6 · HUMIDITY_SPIKE: 12
Detected anomaly events	126 fired by the live detector (window=3 consecutive readings), last at 2026-10-15 12:30:00
Alerts in alerts.csv	33 total · 15 CRITICAL

Counts intentionally differ: labelled readings are dataset ground-truth episodes; breach readings are raw out-of-bounds values; detected events are spawned only after one metric stays out of bounds for 3 consecutive readings.

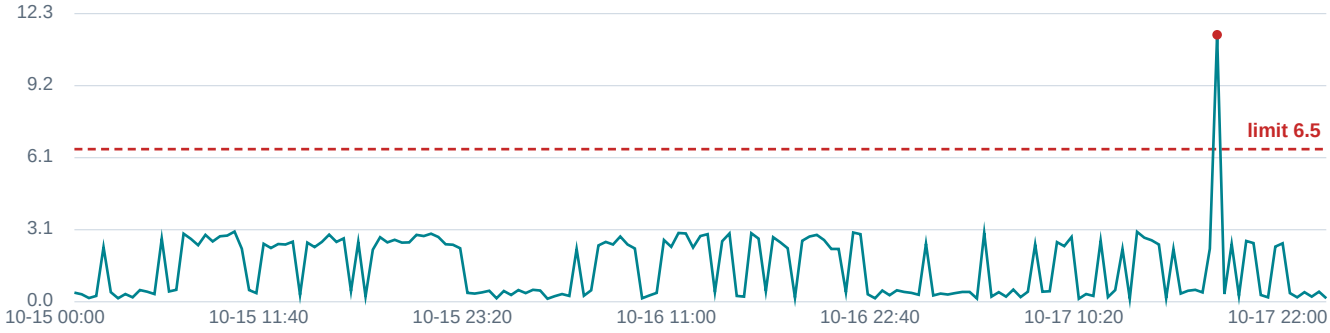
SENSOR PERFORMANCE GRAPHS

TEMPERATURE OVER TIME (°C)



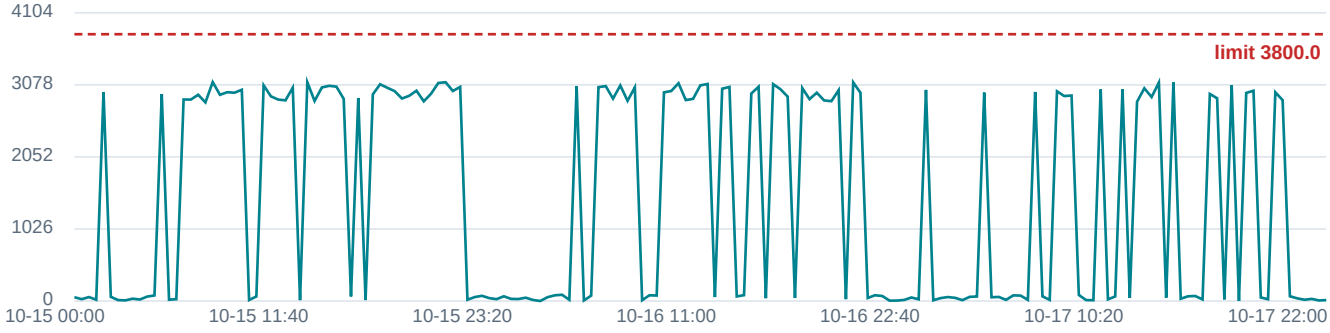
173 plotted readings · red dots = labelled anomaly also breaching this sensor's limit

VIBRATION OVER TIME (MM/S)



173 plotted readings · red dots = labelled anomaly also breaching this sensor's limit

MOTOR RPM OVER TIME (RPM)



173 plotted readings · red dots = labelled anomaly also breaching this sensor's limit

ALERT HISTORY

TIMESTAMP	METRIC	VALUE	THRESHOLD	SEVERITY	MESSAGE
2026-10-17 18:30	humidity_pct	83.3	70.0	WARNING	Humidity above limit
2026-10-17 18:25	humidity_pct	84.2	70.0	WARNING	Humidity above limit
2026-10-17 18:20	humidity_pct	84.7	70.0	WARNING	Humidity above limit
2026-10-17 18:15	humidity_pct	82.1	70.0	WARNING	Humidity above limit
2026-10-17 18:10	humidity_pct	81.7	70.0	WARNING	Humidity above limit
2026-10-17 18:05	humidity_pct	82.2	70.0	WARNING	Humidity above limit
2026-10-17 18:00	humidity_pct	79.8	70.0	WARNING	Humidity above limit
2026-10-17 17:55	humidity_pct	79.6	70.0	WARNING	Humidity above limit
2026-10-17 17:50	humidity_pct	78.4	70.0	WARNING	Humidity above limit
2026-10-17 17:45	humidity_pct	78.9	70.0	WARNING	Humidity above limit
2026-10-17 17:40	humidity_pct	78.0	70.0	WARNING	Humidity above limit
2026-10-17 17:35	humidity_pct	76.8	70.0	WARNING	Humidity above limit
2026-10-17 17:30	vibration_mm_s	11.4	6.5	CRITICAL	Excessive vibration - inspect bearings
2026-10-17 17:30	humidity_pct	78.9	70.0	WARNING	Humidity above limit
2026-10-17 17:25	vibration_mm_s	11.4	6.5	CRITICAL	Excessive vibration - inspect bearings
2026-10-17 17:25	humidity_pct	75.4	70.0	WARNING	Humidity above limit
2026-10-17 17:20	vibration_mm_s	9.8	6.5	CRITICAL	Excessive vibration - inspect bearings
2026-10-17 17:20	humidity_pct	78.9	70.0	WARNING	Humidity above limit
2026-10-17 17:15	vibration_mm_s	9.0	6.5	CRITICAL	Excessive vibration - inspect bearings
2026-10-17 17:15	humidity_pct	75.9	70.0	WARNING	Humidity above limit
2026-10-17 17:10	vibration_mm_s	7.5	6.5	CRITICAL	Excessive vibration - inspect bearings
2026-10-17 17:10	humidity_pct	74.7	70.0	WARNING	Humidity above limit

Showing latest 22 of 33 alerts, newest first. The full history is available in alerts.csv.

SUMMARY & RECOMMENDATIONS

This printer shows a labelled-anomaly rate of 4.1%, in line with the fleet-wide 6.0%.

Vibration was out of bounds on 6 reading(s) (limit 6.5 mm/s, max seen 11.4 mm/s). Recommend inspecting mechanical components such as bearings, mounts and drive belts.

Humidity fell outside the 30–70% band on 18 reading(s). Recommend reviewing site climate control.

Paper jams exceeded 2 per interval on 9 reading(s). Recommend checking feed rollers and media quality.

Note: recommendations are rule-based responses to threshold breaches present in the monitored data. They are maintenance suggestions, not confirmed diagnoses.